

Geldium’s Predictive Insights Report

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1. Summary of Predictive Insights

Our predictive model (XGBoost) identified clear patterns in customer behaviour that correlate with loan delinquency.

The following insights are most relevant for the Collections team:

- High-Risk Segments:** Customers with high credit utilization (>0.6), multiple missed payments (≥4), and low credit scores (0.4) also show elevated risk.
- Key Predictors:** The top three variables influencing delinquency are:
 - Missed_Payments
 - Credit_Utilization
 - Debt_to_Income_Ratio
- Meaningful Patterns:** Customers with zero loan balances but recorded missed payments suggest data inconsistencies or hidden liabilities. These anomalies were flagged during EDA and addressed through imputation and feature engineering.

Key Insights Summary Table

Key Insight	Customer Segment	Influencing Variables	Potential Impact
High credit utilization increases risk	Overextended borrowers	Credit_Utilization, Missed_Payments	Targeted outreach and restructuring options
Low credit score correlates with delinquency	Financially vulnerable individuals	Credit_Score, Employment_Status	Adjust risk thresholds and offer financial counselling
High debt-to-income ratio signals stress	Mid-income earners with multiple loans	Debt_to_Income_Ratio, Monthly Payment Status	Prioritize for proactive repayment support

2. Recommendation Framework

1) Restated Insight:

High credit utilization (>0.6) is a strong predictor of delinquency and affects a large segment of the customer base.

2) Proposed Recommendation:

Launch a proactive engagement program targeting customers with high credit utilization to reduce delinquency risk.

3) Specific:

Identify and engage customers with Credit_Utilization > 0.6 through personalized financial counselling and repayment restructuring options.

4) Measurable:

Achieve a 15% reduction in delinquency among this segment within one quarter.

5) Actionable:

Use existing CRM and analytics tools to flag high-risk customers and automate outreach workflows.

6) Relevant:

Supports Geldium's strategic goal of reducing default rates and improving repayment behaviour.

7) Time-bound:

Roll out the program by Q4 FY25, with monthly performance tracking and adjustments.

8) Justification and Business Rationale:

This recommendation is backed by model insights and aligns with operational feasibility. By addressing overextension early, Geldium can reduce losses, improve customer outcomes, and demonstrate responsible lending practices.

3. Ethical and Responsible AI Considerations

- **Fairness Risks:**

- The model may disproportionately flag unemployed or low-income customers.

Mitigation: Apply fairness-aware algorithms and monitor disparate impact metrics.

- Historical bias in credit scoring could reinforce systemic disadvantages.

Mitigation: Reweight training data and conduct regular fairness audits.

- **Explainability:**

Model predictions are communicated using feature importance visuals and plain language summaries (e.g., "Your credit usage and missed payments suggest elevated risk").

- **Responsible Decision-Making:**

The recommendation promotes early intervention rather than punitive action, supporting financial health and customer retention.

- **Ethical Principles Considered:**

Transparency (documented assumptions), accountability (human-in-the-loop review), and data privacy (secure handling of customer data) are embedded throughout the process.